EXCEL ADD-IN · 17-PAGE OFFICIAL ENGLISH GUIDE

SniperSheet

AI-Powered Excel Formula Engine

Natural Language Precise Excel Formulas in Seconds

35+	4	100%	3	2
Formula Patterns	Smart Tabs	Free AI	AI Models	Languages

Smart Hub · Commands · Cell Dimensions · Advanced Tools · Local Fallback · 100% Free · Bilingual

Developed by: Mustafa Alsahlany

Table of Contents

Complete 17-page guide — click any section to navigate

What is SniperSheet?

AI-powered Excel add-in for natural language formula generation

SniperSheet is a professional Microsoft Excel task pane add-in that transforms natural language into precise, ready-to-use Excel formulas. Powered by Groq AI (Llama 3.3 70B — completely free), it supports both English and Arabic input, routes all AI calls through a Replit US server (no VPN anywhere in the world), and falls back automatically to a built-in local formula engine when offline.

Groq AI Engine
Llama 3.3 70B
100% free — forever

Bilingual Support
English & Arabic input
Full RTL layout

Proxy Architecture
US Replit server
No VPN needed

Local Fallback
35+ instant patterns
Zero latency
Works offline

Office.js Add-in
ReadWriteDocument
Manifest v1.1
Excel Ribbon tab

400px Task Pane
Smooth scroll UX
Native Excel
Mobile compatible

Describe your calculation in plain English or Arabic ▯ SniperSheet generates the exact formula with

Who is SniperSheet For?

Target professionals and their primary use cases

Engineers
Complex mechanical, electrical & civil engineering

Financial Analysts
KPI tracking, budget analysis, and financial modeling

Project Managers
Progress tracking, scheduling, and resource

Students & Academics
Statistical analysis, research data, and grade calculations

Operations Teams
Inventory management, capacity planning, logistics

Sales & Marketing
Commission calculations, pipeline analysis, ROI tracking

Task Pane Interface

400px width · 4-tab navigation · bilingual RTL/LTR support

SniperSheet operates as a 400px-wide task pane attached to the right side of Microsoft Excel. The interface is built in React 18 + Vite and styled with Tailwind CSS. It supports both Left-to-Right (English) and Right-to-Left (Arabic) layouts dynamically, with smooth scrolling within each section, native Office.js integration, and a fe

01 Smart Hub

- AI formula generation via Groq
- Confidence score (0–100%)
- Formula type auto-detection
- Word Radar for typo detection
- Style hints & formatting detection
- History log with timestamps
- Example prompts library

02 Commands

- Arabic alias commands
- SUM, AVERAGE, MAX, MIN...
- BONUS (15%), TAX (15%)
- Real-time calculation
- Searchable reference table
- Execution history

03 Cell Dimensions

- Arabic-aware width calc
- Font size & bold awareness
- Single & batch modes
- Pixel and Excel units
- Recommended padding
- One-click copy results

04 Advanced Tools

- Empty Field Radar scan
- Smart Print-Fit calculator
- Professional Report view
- CSV / tab-data input
- Color-coded grid output
- Browser print dialog

Design System

Color palette, typography, and visual language

Name	Hex
Primary Green	#107C41
Dark Green	#0B5C30
Light Green	#E8F5EE
Neutral 900	#111827
Neutral 600	#4B5563
Neutral 200	#DDE3EA

Smart Hub — AI Formula Engine

Tab 1 · Natural language description ▯ Excel formula via Groq AI

The core feature of SniperSheet. Type any calculation description in English or Arabic — such as "calculate overtime for hours over 40 at 1.5x rate" — then click "Smart Analysis". The AI engine routes your request through Groq AI (Llama 3.3 70B) and returns the exact formula, a full step-by-step explanation, a confidence
e (0–100%), and optional style hints.

Complete Smart Hub Feature List

Every capability of the AI formula engine

- AI formula generation (Groq Llama 3.3 70B)
- Confidence scoring: 0–100% per result
- Formula type auto-detection (IF, XLOOKUP, SUM...)
- Word Radar: detects typos & ambiguous terms
- Style hints: cell color, bold, italic formatting
- Full history log with timestamps & replay
- Example prompts library (English & Arabic)
- Keyboard shortcut: Ctrl+Enter for quick analysis
- Local formula engine — fully offline fallback
- 3-model AI cascade with auto-retry on failure
- Status badge: success / warning / error
- Detailed step-by-step formula explanation
- One-click copy formula to clipboard
- One-click apply formula to active Excel cell
- Supports deeply nested complex formulas
- Smart error messages with suggested fixes

Smart Hub Workflow

Step-by-step: how a formula is generated

1 **Type**
Write your formula description in plain English or Arabic in

2 **Analyze**
Click "Smart Analysis" or press Ctrl+Enter to trigger

3 **Route**
Request goes via HTTPS to Replit server, which calls

4 **Return**
Formula + explanation + confidence score

5 **Apply**
Click "Copy" or "Apply to Cell" to use the formula directly in

Pro Tip: The more specific your description, the higher the confidence score. Include column references (e.g., "column A"), conditions (e.g., "greater than 1000"), and the expected output type. Try the built-in example prompts as a starting point.

Commands — Formula Aliases

Tab 2 · Arabic & English shorthand for instant formula execution

Type the command name in English or Arabic along with a cell range to get an instant result. The Commands tab provides a searchable reference table of all available function aliases, real-time calculation display, and a full execution history. Arabic aliases make formula entry natural for Arabic-speaking users.

Command (EN / AR)	Description	Formula / Output
SUM / جمع	Sum of all values in range	=SUM(B1:B10)
MULTIPLY / ضرب	Product (multiply) of values	=PRODUCT(A1:A5)
AVERAGE / متوسط	Arithmetic mean of range	=AVERAGE(D1:D20)
MAX / أكبر	Maximum value in range	=MAX(C1:C50)
MIN / أصغر	Minimum value in range	=MIN(C1:C50)
COUNT / عدد	Count of numeric cells	=COUNT(E1:E100)
BONUS / مكافأة	Adds 15% bonus to amount	=A1*1.15
TAX / ضريبة	Calculates 15% tax on amount	=A1*0.15
PERCENTAGE / نسبة	Percentage of value vs total	=(A1/B1)*100
IF / إذا	Simple conditional logic check	=IF(A1>1000,"High","Low")
COUNTIF / شرطعدد	Count cells matching condition	=COUNTIF(B:B,">=60")
SUMIF / شرطجمع	Sum values matching condition	=SUMIF(C:C,">1000",D:D)

Supported Features

- Searchable command reference table
- Real-time result calculation
- Command execution history with timestamps
- Combine commands (e.g. BONUS + SUM)

Usage Examples

```
Type: "SUM B1:B10" !′ =SUM(B1:B10)
Type: "BONUS A1" !′ =A1*1.15
Type: "bÆEc′ # ¤## "  " Õ5TÒ„# ¤## •
Type: "PERCENTAGE A1 B1" !′ =(A1/B1)*100
```

Cell Dimensions Calculator

Tab 3 · Optimal cell sizing for Arabic and English text

Calculates the optimal width and height for Excel cells based on content, accounting for Arabic character width differences, RTL text direction, font size, bold and italic styling, and padding margins. Eliminates the guesswork of manually resizing cells for bilingual spreadsheets. Supports single-cell and batch processing es.

How the Calculator Works

Input parameters ▯ dimension algorithm ▯ Excel units output

Text Content

Enter the text that will go inside the cell. Supports Arabic, English, or mixed content. The algorithm Arabic characters lies the correct width multiplier.

Font Size (pt)

Specify the font size in points. The calculator adjusts both width (character px ÷ 7.5) and height (line-height × rows) based on the font size you provide.

Bold & Italic

Toggle bold or italic to apply the corresponding width multiplier (bold: ×1.15, italic: ×1.05). Combinations are calculated cumulatively.

Padding Margin

Specify padding in pixels on each side. Added to the final calculated width and height. Default: 8px horizontal, 4px vertical per cell.

Output Formats & Features

What the calculator returns and available actions

- Width result in pixels (px)
- Width result in Excel column units
- Height result in pixels (px)
- Height result in Excel row units
- Recommended minimum padding values
- Character count & text metrics

- Single cell calculation mode
- Batch processing: multiple cells at once
- One-click copy results to clipboard
- Apply dimensions directly to active Excel cell
- Save and recall favorite settings
- Arabic / English / mixed content aware

Arabic Width Insight: Arabic characters are on average 40% wider than Latin characters at the same font size. The Cell Dimensions calculator applies a 1.4× Arabic character width multiplier automatically, ensuring Arabic text never overflows its cell.

Advanced Tools — 3 Professional Utilities

Tab 4 · Empty Radar, Print-Fit, and Professional Report

Empty Field Radar

Tab 4 · Feature

Scan any dataset for missing values with a color-coded visual grid

Paste CSV or tab-separated data into the input area, click "Scan", and instantly get a color-coded grid showing filled (green) and empty (red) cells, plus a precise list of every empty cell coordinate. Perfect for data validation before analysis.

- Paste CSV or tab-separated data input
- Visual grid: green=filled, red=empty
- Exact empty cell coordinate list (e.g. Row3:Col2)
- Auto header row detection
- Filled vs. empty count summary

Smart Print-Fit

Tab 4 · Feature

Calculate exact column widths and font sizes for A4/A3 paper

Input your column count, row count, and target paper size (A4 or A3) in portrait or landscape orientation. Smart Print-Fit calculates the exact Excel column width and font size needed so your data fits perfectly on one page — no trial and error.

- A4 and A3 paper size support
- Portrait and landscape orientation modes
- Precise Excel column width unit output
- Recommended font size for row count
- Margin and padding adjustment controls

Professional Report View

Tab 4 · Feature

Transform raw pasted data into a styled printable HTML report

Paste any raw data (CSV or tab-separated) and Professional Report View renders it as a styled HTML table with alternating row colors, bold column headers, and a professional layout — then triggers the browser's print dialog for immediate printing.

- Auto-styled HTML table from raw data
- Alternating row color scheme (white/grey)
- Bold column headers with accent color
- Auto header row detection and formatting
- One-click browser print dialog trigger
- Saves paper with optimal table layout

AI Proxy Architecture

How SniperSheet securely connects users to Groq AI via Replit

SniperSheet uses a server-side proxy architecture to securely relay AI requests. The GROQ_API_KEY is stored as a secret environment variable on the Replit server — it is never exposed to the browser or the user. This means users anywhere in the world, including Iraq, connect only to the Replit HTTPS endpoint and …PI. No VPN, … d is required.

User Browser
Excel Add-in
React 18 + Vite
HTTP POST /api/smart

HTTPS
port 443 / Request
TLS

Replit Server
Express.js Proxy
Node.js 24 · US
API key hidden

Groq SDK
Official SDK

Groq AI Cloud
Llama 3.3 70B
30 RPM Free
JSON response

Security Model

The GROQ_API_KEY is a Replit server secret — stored in the environment, never in client code. The browser only sends the formula description text. Users never see or interact with the Groq API directly. HTTPS/TLS encryption protects all traffic.

No-VPN Design Rationale

Why users in Iraq and restricted regions can access SniperSheet

Traditional Approach (VPN Required)

- User in Iraq needs VPN software
- VPN exposes user's traffic to VPN provider
- VPN cost or performance overhead
- VPN blocked by some networks
- Complex setup — barriers to adoption

SniperSheet Approach (No VPN Needed)

- User connects to Replit.dev domain
- Replit server calls Groq on their behalf
- Replit is accessible globally via HTTPS
- No Groq API exposure to client
- Simple — just install the add-in and use

Groq AI

The AI engine powering SniperSheet — 100% free, no credit card required

FREE FOREVER
No billing · No API key for users

3-Model Cascade System

How SniperSheet automatically tries models in order

PRIMARY

llama-3.3-70b-versatile

Params	Ctx	RPM
70B	32768 tok	30

Highest quality · Latest Llama 3.3 release · Best for complex formulas · Low latency [illegible]'s LPU hardware

FALLBACK

llama3-70b-8192

Params	Ctx	RPM
70B	8192 tok	30

Proven stable · Identical parameter count · Smaller context window (8K tokens) · Activates if primary is rate-limited

EMERGENCY

llama3-8b-8192

Params	Ctx	RPM
8B	8192 tok	30

Ultra-fast inference · Lightweight 8B parameter model · For simple formulas · Activates if first two models fail

OFFLINE

Local Formula Engine

Params	Ctx	RPM
—	— tok	∞

Zero latency · No network needed · 35+ built-in patterns · Always available · Activates when all cloud models fail

Groq LPU Technology

Why Groq AI is extremely fast

Groq runs on its own Language Processing Unit (LPU) hardware — purpose-built silicon for AI inference. Unlike GPU-based systems, LPUs deliver consistent, ultra-low latency without batching delays. SniperSheet users typically receive their formula result within 1–3 seconds, even for complex nested formulas with

~1-3s	30 RPM	0$	100%
Formula Response	Free Rate Limit	Cost to Users	Availability

Free Tier Summary: Groq provides 30 requests/minute at no cost for all three Llama models listed above. SniperSheet is designed to stay within this limit. The cascade system ensures 99%+ formula generation success even during peak usage.

Replit

The cloud platform hosting the SniperSheet API server — Autoscale deployment on US infrastructure

AUTOSCALE
Always-on · US region

Server Stack & Configuration

Complete technical profile of the SniperSheet backend

Property	Value	Notes
Runtime	Node.js 24 (LTS)	Latest LTS with full ESM support and improved
Framework	Express.js 5	Modern async/await middleware, improved error handling
AI SDK	Groq SDK (Official)	Official groq-sdk npm package — TypeScript native
Language	TypeScript (strict)	Full type safety across all route handlers and middleware
Build Tool	esbuild	Fast CJS bundle output for production deployment
Pkg Manager	pnpm Workspace	Monorepo-compatible, fast installs, disk-efficient
Platform	Replit Autoscale	Auto-scales to zero when idle, spins up on demand
Region	United States (US)	Low latency to Groq AI cloud · accessible globally
Protocol	HTTPS / TLS 1.3	All traffic encrypted · picard.replit.dev domain
CORS Policy	Origin-restricted (HTTPS)	Add-in domain whitelisted · browser security enforced

API Endpoints

Routes served by the SniperSheet Express server

Method	Route	Description
POST	`/api/smart/analyze`	Main AI formula generation endpoint — accepts {prompt, lang}
GET	`/api/addin/manifest.xml`	Downloads Office.js manifest for Excel add-in sideloading
GET	`/api/addin/icon-*.png`	Serves PNG icons: 16, 32, 64, 80px sizes for the ribbon
GET	`/excel-addin/`	Serves the React task pane SPA (Vite production build)

Real Formula Examples

12 formulas generated by Smart Hub AI from natural language

Natural Language Input	Formula Generated	Category
Calculate overtime: hours over 40, rate 1.5× hourly wage	`=IF(A1>40,(A1-40)*1.5,0)`	Conditional
Grade student: A+ above 90, B above 75, C otherwise	`=IFS(B1>=90,"A+",B1>=75,"B","C")`	Conditional
Add 15% bonus if employee sales exceed 10,000	`=IF(A1>10000,A1*1.15,A1)`	Conditional
Find value in column A, return matching from column B	`=XLOOKUP(D1,A:A,B:B,"Not found")`	Lookup
Look up product price from price list table	`=VLOOKUP(A2,Table1,3,0)`	Lookup
Sum all sales greater than 1000 in column D	`=SUMIF(C:C,">1000",D:D)`	Statistical
Count students who scored 60 or higher	`=COUNTIF(B:B,">=60")`	Statistical
Rank each employee by highest sales (descending)	`=RANK(A1,A:A,0)`	Statistical
Monthly payment for 30-year loan at 5% annual interest	`=PMT(0.05/12,360,A1)`	Financial
Calculate compound interest over 10 years at 7%	`=A1*(1+0.07)^10`	Financial
Calculate person's age in years from birth date in A1	`=DATEDIF(A1,TODAY(),"Y")`	Date
Count working days between two dates (exclude weekends)	`=NETWORKDAYS(A1,B1)`	Date

Category Key: Conditional · Lookup · Statistical · Financial · Date

Confidence: Green badge (>85%) = high accuracy · Orange (50–85%) = acceptable, try to be more specific · Red (<50%) = revise your prompt with more details and column references. Clearer prompts always yield better results.

Writing Effective Formula Prompts

How to describe your calculation for the best AI results

The quality of your formula prompt directly determines the accuracy and confidence of the AI result. Effective prompts are specific, include column or cell references, define conditions clearly, and state the expected output. This page shows you exactly how to write prompts that consistently produce 90%+ confidence scores.

Prompts That Work Well

Examples of effective formula descriptions

Calculate overtime pay: if hours worked in A1 exceed 40, multiply the excess by 1.5 times the hourly rate in B1

Score: 96%

Why it works: Specifies the cells (A1, B1), the condition (> 40), and the calculation method (1.5× rate)

!’ =IF(A1>40,(A1-40)*B1*1.5,0)

Look up the employee ID in column A and return their monthly salary from column C in the same row

Score: 94%

Why it works: Clearly names both columns, defines the lookup key, and specifies the return column

!’ =XLOOKUP(F2,A:A,C:C,"Not found")

Sum all values in column D where the corresponding value in column C is greater than 1000

Score: 97%

Why it works: Names both columns, defines the condition column and value column separately

!’ =SUMIF(C:C,">1000",D:D)

Grade a student: if the score in B2 is 90 or above give A+, if 75 or above give B, otherwise give C

Score: 99%

Why it works: States all grade boundaries explicitly and in the correct order for IFS function

!’ =IFS(B2>=90,"A+",B2>=75,"B",TRUE,"C")

Prompts to Avoid

Vague descriptions that produce low-confidence results

Vague: "Calculate something with numbers in column A"

Fix: Be specific — name the operation (SUM, IF, AVERAGE) and what numbers represent

Vague: "Make a formula for employee data"

Fix: Describe the exact calculation — e.g., "sum salaries if department equals Sales"

Vague: "Do a lookup thing"

Fix: Specify: what to look up, where to find it, and what to return — then name the columns

Vague: "Calculate tax"

Fix: State the rate: "calculate 15% tax on the amount in cell A1" gives much better results

Confidence Score System

How SniperSheet measures and displays formula accuracy

Every formula generated by the Smart Hub AI engine is accompanied by a confidence score from 0 to 100%. This score is calculated by the AI model based on how clearly your prompt described the required calculation, how well the generated formula matches the description, and the complexity of the formula pattern used.

85–100%

High Confidence

The formula is highly likely to be correct. The prompt was clear and specific. Use this formula as-is with confidence.

- Prompt named specific columns or cells
- Condition values were stated explicitly
- Formula type was clearly implied by the description

`Score 97%: "sum sales in D where C > 1000" !’ =SUMIF(C:C,">1000",D:D)`

50–84%

Acceptable

The formula is likely correct but review it before using. The prompt had some ambiguity or missing details.

- Prompt was partially specific
- Condition or column was implied, not stated
- Formula type was inferred from context

`Score 71%: "calculate bonus for sales" !’ =A1*1.15 (assumed 15% bonus)`

0–49%

Low Confidence

The formula may be incorrect. Revise your prompt with more specific details, column names, and condition values.

- Prompt was too vague or general
- No specific columns, values, or conditions mentioned
- Multiple possible interpretations existed
- Word Radar detected unclear or ambiguous terms

`Score 34%: "do something with data" !’ best guess formula generated`

Word Radar is a built-in prompt analysis tool that runs before the AI call. It detects potentially ambiguous terms, possible typos, and unclear references — then shows a yellow warning with suggestions to improve your prompt before submitting.

Installation Guide

4 steps to load SniperSheet in Microsoft Excel

1

Download manifest.xml

Open your browser and navigate to the manifest URL:

https://[your-replit-domain]/api/addin/manifest.xml

2

Open Excel Add-ins Manager

In Microsoft Excel: click the Insert tab in the Ribbon click Add-ins click My Add-ins click "Manage My Add-ins" at the bottom of the dropdown.

3

Upload manifest.xml

In the Add-ins Manager: click "Upload My Add-in" select the manifest.xml file you downloaded. The SniperSheet tab will appear automatically in the Excel Ribbon.

4

Open Smart Hub

Click the SniperSheet tab in the Excel Ribbon click "Open Sniper Hub". The task pane opens on the right side of Excel. You're ready to generate formulas with AI!

System Requirements

Minimum requirements to run SniperSheet

Requirement	Value	Notes
Operating System	Windows 10/11 or macOS 12+	Required for Office.js task pane add-ins
Microsoft Excel	Excel 2016 or later	Office 365 recommended for full features
Internet	Required for AI features	Offline local engine works without internet
Browser Engine	Edge WebView2 or Chrome	Embedded browser handles the task pane UI
Screen	1280×720 or higher	Task pane requires minimum 400px sidebar space

 Manifest Download URL (copy this into your browser):

`https://8e832e48-8f9e-4168-9828-29c19ce7accc-00-12f81e1kjeof1.picard.replit.dev/api/addin/manifest.xml`

The server generates the manifest dynamically — always current with the latest version

Frequently Asked Questions

8 common questions about SniperSheet answered

Is SniperSheet 100% free to use?

Yes. SniperSheet uses Groq's free tier (Llama 3.3 70B, 30 requests/minute). The GROQ_API_KEY is managed by the developer on the server. Users never need to create a Groq

Do users in Iraq need a VPN?

No. All AI requests are routed through a Replit server hosted in the United States. Users connect to the Replit HTTPS endpoint, not directly to Groq. This proxy architecture means users in

What happens when the AI is unavailable or rate-limited?

SniperSheet uses a 3-model cascade: if the primary model (Llama 3.3 70B) is rate-limited or fails, it automatically retries with the second model (Llama 3 70B 8192), then the third

Does SniperSheet work without internet?

Yes — partially. The local formula engine works 100% offline and handles SUM, IF, AVERAGE, VLOOKUP, and 35+ other common patterns with no network connection. The AI Smart

What Excel versions are supported?

Microsoft Excel 2016 and later on Windows and Mac. Office 365 is recommended for the best experience. The add-in uses the Office.js Manifest v1.1 specification with

Can I type my formula request in Arabic?

Absolutely. Arabic language input is fully supported in Smart Hub. Type your description in Arabic, and SniperSheet generates the exact Excel formula along with a full explanation

How secure is my formula data?

Your formula descriptions are sent over HTTPS to the Replit server, which forwards only the text prompt to Groq AI. No user data is stored persistently. The local history log exists only in the browser session memory and is cleared when you close the task pane.

Can SniperSheet write formulas directly into Excel cells?

Yes. With ReadWriteDocument permission, SniperSheet can write formulas directly into the currently active Excel cell using the Office.js API. Click "Apply to Cell" after a formula is generated to insert it. You can also manually copy and paste the formula.

Developer Profile

The creator of SniperSheet

Mustafa Alsahlany

Developer & Designer — SniperSheet Excel Add-in

Built with React, Express.js, TypeScript, and Groq AI on Replit

Iraq · 2026

- Replit — Active
- Groq AI — Free
- v1.0.0
- April 2026

Tech Stack

- Frontend: React 18 + Vite + TypeScript
- Backend: Express.js + Node.js 24
- AI Engine: Groq SDK — Llama 3.3 70B
- Styling: Tailwind CSS + shadcn/ui
- Packages: pnpm Monorepo Workspace
- Build: esbuild (CJS bundle)

Tech Stack

- Manifest: Office.js v1.1 Task Pane
- Permission: ReadWriteDocument
- Locale: ar-SA (primary) + en-US
- Hosting: Replit Autoscale (US)
- Security: HTTPS / TLS 1.3
- Icons: 16 · 32 · 64 · 80 px PNG

Groq AI

Llama 3.3 70B · Free Tier · 30 RPM

3-Model Cascade · LPU Hardware

AI engine — no billing — no account needed

Replit

Autoscale · US Region · HTTPS

Node.js 24 · Express.js · API Proxy

No VPN needed — accessible worldwide

SniperSheet v1.0 · Official English Guide · 17 Pages · Built on Replit · Powered by Groq AI (Free Tier) · Developed by: Mustafa Alsahlany